

Forensic Detection Report

Report: FR-MULTI-20260930T081238Z-1f05dcb1
Generated (UTC): 2026-09-30T08:12:38.681394+00:00

This is a system-generated report. All detections and validation results are subject to human analyst review and should be treated as reviewable observations, not conclusive findings.

Video Metadata

CAM-01 - SYNTHETIC_CAM-01.mp4

Video filename	SYNTHETIC_CAM-01.mp4
Resolution	1280 x 720 pixels
Frame rate	10 FPS
Duration	100.000 seconds
Source frame count	1000
Camera identifier	CAM-01
Recording date/time	Not recorded

CAM-02 - SYNTHETIC_CAM-02.mp4

Video filename	SYNTHETIC_CAM-02.mp4
Resolution	1280 x 720 pixels
Frame rate	10 FPS
Duration	100.000 seconds
Source frame count	1000
Camera identifier	CAM-02
Recording date/time	Not recorded

Video and Detection Information

CAM-01 - SYNTHETIC_CAM-01.mp4

Model used	SYNTHETIC_MODEL.pth
Run status	Completed saved run
Confidence threshold	50%
Frames analyzed	1000
Processing device / time	CPU / 1.00 seconds
Analysis completed (UTC)	2026-09-30T08:12:38+00:00
Detection summary	48 observations; 48 positive frames; Knife: 48
Analyst review coverage	1 of 48 observations reviewed

CAM-02 - SYNTHETIC_CAM-02.mp4

Model used	SYNTHETIC_MODEL.pth
Run status	Completed saved run
Confidence threshold	50%
Frames analyzed	1000
Processing device / time	CPU / 1.00 seconds
Analysis completed (UTC)	2026-09-30T08:12:38+00:00
Detection summary	2 observations; 2 positive frames; Knife: 2
Analyst review coverage	0 of 2 observations reviewed

Interpretation

The model generated the listed handgun and knife observations at the configured confidence threshold. A confidence score describes model certainty and is not an analyst decision. Analyst decisions document a later human assessment and do not replace or delete the original model observation.

Frame numbers are zero-based. Times are video-relative offsets and are not recording dates. Bounding boxes use [x1, y1, x2, y2] source-frame pixel coordinates. Counts represent frame observations, so the same physical object may appear more than once.

Model Performance Metrics

CAM-01 - SYNTHETIC_CAM-01.mp4

Metric	Temporal Consistency Rate (TCR)
Result	N/A
Interpretation	Unfiltered pipeline records were not saved for this run. Process the video again to calculate this metric.

CAM-02 - SYNTHETIC_CAM-02.mp4

Metric	Temporal Consistency Rate (TCR)
Result	N/A
Interpretation	Unfiltered pipeline records were not saved for this run. Process the video again to calculate this metric.
Metric	Multi-Camera Corroboration Rate (MCCR)
Result	0.00%
Interpretation	Calculated from aligned observations across the imported camera recordings.
Compared cameras	CAM-01 and CAM-02
Corroborated observations (N_CC)	0
Not corroborated	48
Uncertain / conflicting labels	0
Outside concurrent window	2
Eligible observations (N_MC)	48
Concurrent comparison window	Matching window +/- 0.25 seconds

Object Detection Observations and Reviews

Each row is a frame observation. Confidence and cross-view matching remain reviewable. Session time is video time plus the camera offset; it is not a recording date/time. The CSV and JSON files contain all record identifiers, matching details and source references.

Session time	Camera	Video time	Frame	Object	Confidence	Box [x1,y1,x2,y2]	Cross-view
0.000 s	CAM-02	0.000 s	0	Knife	91.23%	[0, 20, 10, 40]	Not Applicable
0.100 s	CAM-02	0.100 s	1	Knife	91.23%	[1, 20, 11, 40]	Not Applicable
2.000 s	CAM-01	0.000 s	0	Knife	91.23%	[0, 20, 10, 40]	Not Corroborated
2.100 s	CAM-01	0.100 s	1	Knife	91.23%	[1, 20, 11, 40]	Not Corroborated
2.200 s	CAM-01	0.200 s	2	Knife	91.23%	[2, 20, 12, 40]	Not Corroborated
2.300 s	CAM-01	0.300 s	3	Knife	91.23%	[3, 20, 13, 40]	Not Corroborated
2.400 s	CAM-01	0.400 s	4	Knife	91.23%	[4, 20, 14, 40]	Not Corroborated
2.500 s	CAM-01	0.500 s	5	Knife	91.23%	[5, 20, 15, 40]	Not Corroborated
2.600 s	CAM-01	0.600 s	6	Knife	91.23%	[6, 20, 16, 40]	Not Corroborated
2.700 s	CAM-01	0.700 s	7	Knife	91.23%	[7, 20, 17, 40]	Not Corroborated
2.800 s	CAM-01	0.800 s	8	Knife	91.23%	[8, 20, 18, 40]	Not Corroborated
2.900 s	CAM-01	0.900 s	9	Knife	91.23%	[9, 20, 19, 40]	Not Corroborated
3.000 s	CAM-01	1.000 s	10	Knife	91.23%	[10, 20, 20, 40]	Not Corroborated
3.100 s	CAM-01	1.100 s	11	Knife	91.23%	[11, 20, 21, 40]	Not Corroborated
3.200 s	CAM-01	1.200 s	12	Knife	91.23%	[12, 20, 22, 40]	Not Corroborated
3.300 s	CAM-01	1.300 s	13	Knife	91.23%	[13, 20, 23, 40]	Not Corroborated
3.400 s	CAM-01	1.400 s	14	Knife	91.23%	[14, 20, 24, 40]	Not Corroborated
3.500 s	CAM-01	1.500 s	15	Knife	91.23%	[15, 20, 25, 40]	Not Corroborated

Session time	Camera	Video time	Frame	Object	Confidence	Box [x1,y1,x2,y2]	Cross-view
3.600 s	CAM-01	1.600 s	16	Knife	91.23%	[16, 20, 26, 40]	Not Corroborated
3.700 s	CAM-01	1.700 s	17	Knife	91.23%	[17, 20, 27, 40]	Not Corroborated
3.800 s	CAM-01	1.800 s	18	Knife	91.23%	[18, 20, 28, 40]	Not Corroborated
3.900 s	CAM-01	1.900 s	19	Knife	91.23%	[19, 20, 29, 40]	Not Corroborated
4.000 s	CAM-01	2.000 s	20	Knife	91.23%	[20, 20, 30, 40]	Not Corroborated
4.100 s	CAM-01	2.100 s	21	Knife	91.23%	[21, 20, 31, 40]	Not Corroborated
4.200 s	CAM-01	2.200 s	22	Knife	91.23%	[22, 20, 32, 40]	Not Corroborated
4.300 s	CAM-01	2.300 s	23	Knife	91.23%	[23, 20, 33, 40]	Not Corroborated
4.400 s	CAM-01	2.400 s	24	Knife	91.23%	[24, 20, 34, 40]	Not Corroborated
4.500 s	CAM-01	2.500 s	25	Knife	91.23%	[25, 20, 35, 40]	Not Corroborated
4.600 s	CAM-01	2.600 s	26	Knife	91.23%	[26, 20, 36, 40]	Not Corroborated
4.700 s	CAM-01	2.700 s	27	Knife	91.23%	[27, 20, 37, 40]	Not Corroborated
4.800 s	CAM-01	2.800 s	28	Knife	91.23%	[28, 20, 38, 40]	Not Corroborated
4.900 s	CAM-01	2.900 s	29	Knife	91.23%	[29, 20, 39, 40]	Not Corroborated
5.000 s	CAM-01	3.000 s	30	Knife	91.23%	[30, 20, 40, 40]	Not Corroborated
5.100 s	CAM-01	3.100 s	31	Knife	91.23%	[31, 20, 41, 40]	Not Corroborated
5.200 s	CAM-01	3.200 s	32	Knife	91.23%	[32, 20, 42, 40]	Not Corroborated
5.300 s	CAM-01	3.300 s	33	Knife	91.23%	[33, 20, 43, 40]	Not Corroborated

Session time	Camera	Video time	Frame	Object	Confidence	Box [x1,y1,x2,y2]	Cross-view
5.400 s	CAM-01	3.400 s	34	Knife	91.23%	[34, 20, 44, 40]	Not Corroborated
5.500 s	CAM-01	3.500 s	35	Knife	91.23%	[35, 20, 45, 40]	Not Corroborated
5.600 s	CAM-01	3.600 s	36	Knife	91.23%	[36, 20, 46, 40]	Not Corroborated
5.700 s	CAM-01	3.700 s	37	Knife	91.23%	[37, 20, 47, 40]	Not Corroborated
5.800 s	CAM-01	3.800 s	38	Knife	91.23%	[38, 20, 48, 40]	Not Corroborated
5.900 s	CAM-01	3.900 s	39	Knife	91.23%	[39, 20, 49, 40]	Not Corroborated
6.000 s	CAM-01	4.000 s	40	Knife	91.23%	[40, 20, 50, 40]	Not Corroborated
6.100 s	CAM-01	4.100 s	41	Knife	91.23%	[41, 20, 51, 40]	Not Corroborated
6.200 s	CAM-01	4.200 s	42	Knife	91.23%	[42, 20, 52, 40]	Not Corroborated
6.300 s	CAM-01	4.300 s	43	Knife	91.23%	[43, 20, 53, 40]	Not Corroborated
6.400 s	CAM-01	4.400 s	44	Knife	91.23%	[44, 20, 54, 40]	Not Corroborated
6.500 s	CAM-01	4.500 s	45	Knife	91.23%	[45, 20, 55, 40]	Not Corroborated
6.600 s	CAM-01	4.600 s	46	Knife	91.23%	[46, 20, 56, 40]	Not Corroborated
6.700 s	CAM-01	4.700 s	47	Knife	91.23%	[47, 20, 57, 40]	Not Corroborated

Analyst Review Information

Review coverage	1 of 50 observations reviewed
Decision totals	Accept: 1

Observation ID	Camera / frame	Analyst decision	Analyst notes	Reviewed (UTC)
03657730233c40d98b6e158cb7e6a3b6-OBS-00001	CAM-01 / 0	Accept	Synthetic <review> & reason Second line.	2026-09-30 08:12:38.674098

Source References and Traceability

CAM-01 - SYNTHETIC_CAM-01.mp4

Source video	C:\Users\howar\AppData\Local\Temp\tmpaobwd5b6\CAM-01\SYNTHETIC_CAM-01.mp4
Model	SYNTHETIC_MODEL.pth
Annotated video	C:\Users\howar\AppData\Local\Temp\tmpaobwd5b6\camera-session-20260930-161238-1f05dcb1\SYNTHETIC_CAM-01_20260930_161238_400bbf14\annotated.mp4
Saved summary	C:\Users\howar\AppData\Local\Temp\tmpaobwd5b6\camera-session-20260930-161238-1f05dcb1\SYNTHETIC_CAM-01_20260930_161238_400bbf14\summary.json
Pipeline detections	C:\Users\howar\AppData\Local\Temp\tmpaobwd5b6\camera-session-20260930-161238-1f05dcb1\SYNTHETIC_CAM-01_20260930_161238_400bbf14\detections.csv
Metric input records	C:\Users\howar\AppData\Local\Temp\tmpaobwd5b6\camera-session-20260930-161238-1f05dcb1\SYNTHETIC_CAM-01_20260930_161238_400bbf14\metric_input.csv
Metric calculation script	C:\Users\howar\Documents\GitHub\cctv-weapon-detection-system\benchmarks\calculate_tcr_and_mccr.py

CAM-02 - SYNTHETIC_CAM-02.mp4

Source video	C:\Users\howar\AppData\Local\Temp\tmpaobwd5b6\CAM-02\SYNTHETIC_CAM-02.mp4
Model	SYNTHETIC_MODEL.pth
Annotated video	C:\Users\howar\AppData\Local\Temp\tmpaobwd5b6\camera-session-20260930-161238-1f05dcb1\SYNTHETIC_CAM-02_20260930_161239_f9f1a2de\annotated.mp4
Saved summary	C:\Users\howar\AppData\Local\Temp\tmpaobwd5b6\camera-session-20260930-161238-1f05dcb1\SYNTHETIC_CAM-02_20260930_161239_f9f1a2de\summary.json
Pipeline detections	C:\Users\howar\AppData\Local\Temp\tmpaobwd5b6\camera-session-20260930-161238-1f05dcb1\SYNTHETIC_CAM-02_20260930_161239_f9f1a2de\detections.csv
Metric input records	C:\Users\howar\AppData\Local\Temp\tmpaobwd5b6\camera-session-20260930-161238-1f05dcb1\SYNTHETIC_CAM-02_20260930_161239_f9f1a2de\metric_input.csv
Metric calculation script	C:\Users\howar\Documents\GitHub\cctv-weapon-detection-system\benchmarks\calculate_tcr_and_mccr.py

Session traceability

Report identifier	FR-MULTI-20260930T081238Z-1f05dcb1
Generated (UTC)	2026-09-30T08:12:38.681394+00:00
Camera recordings	2
Incident identifier	TEST-SCENE
Alignment	Confirmed by analyst
Alignment method	Synthetic clock
Matching window	+/- 0.25 seconds

CAM-01 - SYNTHETIC_CAM-01.mp4

Source video fingerprint	2b26ad84a87cb4a2c0f6fd7cfc34f958651b80b62a8535d7918f9442479932d3
Source video check	fingerprinted_at_report_generation
Saved summary fingerprint	519bc6cd0c6a1cc3a9809aa62cc498c4d7be7e1d0f7dd376c58727ed2b0368da
Saved summary check	fingerprinted_at_report_generation
Pipeline detections fingerprint	0161dacc154a1d0c96da97c128a8f3e42c3d6dc1b181de29eff55a57c16373e1
Pipeline detections check	fingerprinted_at_report_generation
Metric input fingerprint	Unavailable
Metric input check	unavailable
Metric script fingerprint	f5124a898945b9d6e3e0ef4b6145a555bcac45faee5b4e92808db933c9539e62

CAM-02 - SYNTHETIC_CAM-02.mp4

Source video fingerprint	2b26ad84a87cb4a2c0f6fd7cfc34f958651b80b62a8535d7918f9442479932d3
Source video check	fingerprinted_at_report_generation
Saved summary fingerprint	63d2f7af324f7e58588e2181707924157e38563a291908ea7c0c9f311a14de89
Saved summary check	fingerprinted_at_report_generation
Pipeline detections fingerprint	1a6ad16b824823960c830b2c297e99b18fef5aec78ae6d13d24d9566821f5954
Pipeline detections check	fingerprinted_at_report_generation
Metric input fingerprint	Unavailable
Metric input check	unavailable
Metric script fingerprint	f5124a898945b9d6e3e0ef4b6145a555bcac45faee5b4e92808db933c9539e62

SHA-256 values identify files read when this report was generated; no ingestion-time hash or chain-of-custody history is asserted.